MIL-E-1/1031C(USAF)
27 April 1972
SUPERSEDING
MIL-E-1/1031B(USAF)
21 November 1958

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 10UP14B

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Magnetic deflection, electrostatic focus, aluminized

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE MAXIMUM RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb1	Eb2
Unit:	V	Vdc	Vdc	Vdc	Vdc
Maximum:	6.9	0	770	1000	13200
Minimum:	5.7	-200	150	-500	7000
TEST CONDITIONS:	6.3	Adjust	300	Adjust	10000

ABSOLUTE MAXIMUM RATINGS:

Parameter:	Ehk	Rg1	Alt
Unit:	Vdc	Meg	ft
Maximum:	180	(See note 1)	15000
Minimum:	-180	---	---
TEST CONDITIONS:	---	---	---

GENERAL:

Qualification - Required

(C) denotes changes

10UP14B

FSC 5960

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:01Z
Check the source to verify that this is the current version before use.

METHOD	TEST OR REQUIREMENT	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Qualification							
1141	Pressure (implosion)		---	---	---	---	---	---
1002	Barometric pressure reduced	440 ±5 Hg	---	---	---	---	---	---
5111	Vibration		---	---	Width	---	2.0	mm
1331	Direct-interelectrode capacitance	k to all g1 to all	--- ---	--- ---	Ck Cg1	--- ---	8 10	pF pF
	Quality conformance inspection, part 1							
5201	Voltage breakdown		0.65	II	---	---	---	---
5201	Voltage breakdown (magnetic types)		0.65	II	---	---	---	---
5206	Gas ratio	See note 2	0.65	II	Gr	---	0.15	---
5101	Neck straightness alinement	cylinder 1.500 + .300 - .000	0.65	II	---	---	---	---
5106	Screen and faceplate blemishes	See note 12	0.65	II	---	---	---	---
5223	Modulation	Ib = 200 µA	0.65	II	ΔEc1	---	42	Vdc
5231	Spot position (magnetic deflection)		0.65	II	---	---	8	mm
5236	Zero-bias anode current (magnetic deflection)		0.65	II	---	---	---	---
5241	Grid-cutoff voltage		0.65	II	Ec1	-40	-70	Vdc
5251	Grid No. 1 leakage		0.65	II	---	---	---	---
5221	Screen brightness II	Ib = 24 µAdc See notes 3 and 6	---	---	D1	450	---	cb
5221	Screen persistence	Ib = 24 µAdc See note7						
----	Blue/orange light ratio	See note 8	---	---	Ratio	1.0	1.5	---
----	Aperture alinement	Eb1 = Eb2 = 5 Kv Ec2 = 300 V See note 14	0.65	II	---	---	8	mm
----	High Impedance Leakage	See note 13	----	---	---	---	---	---
	Quality conformance inspection, part 2							
1101	Secureness of base, cap, or insert	See note 11	---	---	---	---	---	---
1301	Heater current		---	---	If	540	660	mA
5201	Electrode current	Ec1 = 0	---	---	Ic2	-15	15	µAdc
5101	Side terminal and base alinement	Vacant pin position no. 3	---	---	---	---	---	---
5101	Neck and bulb aline-ment (magnetic types)		---	---	---	---	0.30	inch
5101	Face tilt		---	---	---	---	0.10	inch
----	Screen color	See note 4	---	---	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:01Z
Check the source to verify that this is the current version before use.

METHOD	TEST OR REQUIREMENT	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 2 -Continued							
5216	Stray light emission (conventional types)	Eb1 = 0 Eb2 = 13,200 Vdc Ec2 = 770 Vdc	---	---	---	---	---	---
5221	Screen (P14)	See note 3	---	---	---	---	---	---
5221	Screen brightness I	See note 4	---	---	---	---	---	---
5226	Line width A (magnetic deflection)	Ib = 200 µAdc Ib = 60 µAdc See note 9	--- ---	--- ---	Width Width	--- ---	0.68 0.43	mm mm
5226	Line width C (magnetic deflection)	Ib = 200 µAdc Ib = 60 µAdc See note 9	--- ---	--- ---	Width Width	--- ---	0.68 0.43	mm mm
5246	Focusing voltage	Ib = 60 µAdc See note 10	---	---	---	-180	180	Vdc
5251	Heater-cathode leakage current		---	---	---	---	---	---
(S) 1105	Permanence of marking		---	---	---	---	---	---
5251	Grid No. 2 leakage current		---	---	---	---	---	---
(C) 1111	Base pin solder depth		---	---	---	---	---	---
	Quality conformance inspection, part 2							
----	Life-test provisions	Group C t = 500 hours (min) Eb2 = 13,200 Vdc Ec2 = 770 Vdc Ib = 100 µAdc	---	---	---	---	---	---
----	Life-test end points:	Line width A Ib = 60 µAdc	---	---	Width	---	0.43	mm
		Line width C Ib = 60 µAdc	---	---	Width	---	0.43	mm
		Modulation	---	---	ΔEc1	---	42	Vdc

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:01Z
Check the source to verify that this is the current version before use.

NOTES:

1. When Ec2 is greater than 330 Vdc, Rg1 shall not exceed .5 Meg. When Ec2 is less than 330 Vdc, Rg1 shall not exceed 1.5 Meg.

2. This test to be performed at the conclusion of the holding period.

3. The test conditions shall be anode voltage (relative to cathode), 10,000 Vdc, Grid No. 2 300 Vdc.

4. Blue in fluorescence, orange in phosphorescence.

5. Raster shall be repeated 60 times per second.

6. After raster excitation repeated at 1-second intervals until equilibrium has been reached, the light output 1 second after excitation (D1) shall be measured and shall be within the limits specified.

7. After raster excitation repeated at 1-second intervals until equilibrium has been reached, the light output 1 second after excitation shall be 70 to 120 cb less than the light output 0.1 second after excitation. i.e. (D.1-D1) = 70 to 120 cb.

   The drop in light output between 1 and 10 seconds after excitation shall be greater than the drop in light output between .1 and 1 second after excitation for the same tube; i.e. (D1-D10 D.1 -D1).

8. (1) Using tungsten standard luminant A, calibrate Weston Photronic Cell #594-R-R- or equivalent with Corning 4-65 filter or equivalent at 10 foot-candles. This combination approximates the normal eye response.

   (2) Set up a 7.1 cm x 7.1 cm defocused raster on the cathode ray tube under test and adjust the grid-potential for a brightness reading of 10 foot-lamberts when viewed through a Wratten 15 filter by the photronic cell and filter previously standardized.

   (3) Without disturbing the photronic cell, replace Wratten 15 or equivalent and Corning 4-65 filters or equivalent with Corning 5-59 filter or equivalent and again measure the indicated brightness in apparent foot-lamberts. The thickness of the filters shall be 2.95 mm and 5.13 mm respectively.

   (4) Blue/Orange Light Ratio = $\frac{\text{Apparent foot-lamberts as measured with Corning 5-59 filter or equivalent}}{\text{10 foot-Lamberts}}$

9. Use procedure outlined in Note 10 for determination of focus voltage in 0 degree, 60 degree and 120 degree planes. Line width "A" and "C" measurements, made in each of the three planes described, shall not exceed the specified limits in any plane.

10. Use 35 - 105 line raster expanded with approximately 0.50 inch between lines. An individual line at the center shall be examined and focus voltage varied until the line is of minimum width consistent with the absence of background smear. The Eb1 voltage so determined shall be considered the focus voltage and shall be taken in each of three planes (0 degree, 60 degree and 120 degree) through the longitudinal axis of the tube. The 0 degree reference plane is established with the anode button horizontal and to the right. The voltage required for focus in each of the planes shall be within the limit specified and the spread of Eb1 (astigmatism) shall not exceed the specified limit.

11. Soak time shall be 42 hours.

12. There shall be no blemishes greater than 0.015 inch diameter within a 2 inch diameter circle whose center coincides with the undeflected spot position. There shall be no more than 3 blemishes of diameter .010 inch to .015 inch within the same 2 inch diameter circle. Between the outer edge of the 2 inch diameter circle and a 9 inch diameter circle there shall be no blemishes greater than 0.030 inch and no more than 3 between .025 inch to .030 inch diameter. There shall be no blemishes closer than 1/2 inch apart.

Source: https://assist.dla.mil -- Downloaded: 202610-15T14:01Z
Check the source to verify that this is the current version before use.

NOTES: -Continued

13. With the tube mounted face up and the neck accessible, preheat the tube for 2 minutes with the voltages applied as shown in figure 2. Using the mallet and stroke specified in 4.7.5 (MIL-E-1), tap the tube neck and base three times in each of three planes 90 degrees apart (2 normal to the axis and 1 in line with the axis). The meters shall not indicate shorts. GI meter shall never exceed 15 µAdc. K meter shall never exceed 30 µAdc.

14. Vary Ec1 until an aperture circle is completely visible, then reduce Ec1 until the defocused cutoff spot is seen. Evaluate the position of the cutoff spot in relation to the center of the aperture circle. The spot shall be within three fourths of the radius of the aperture circle.

Custodians:
Air Force - 80

Review activities:
Air Force - 11, 17

Preparing activity:
Air Force - 80

(Project 5960-E896)

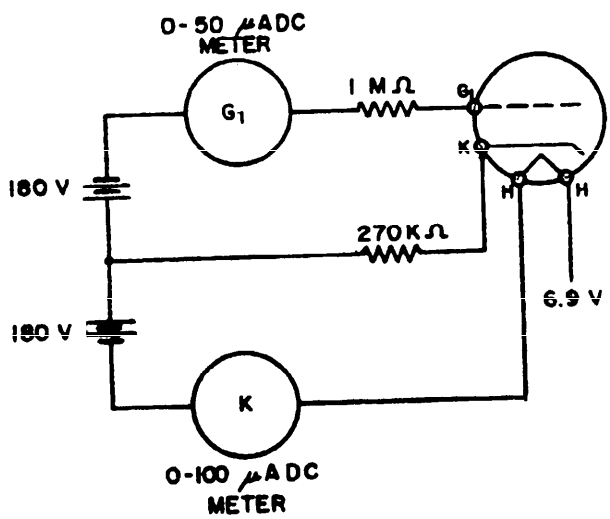
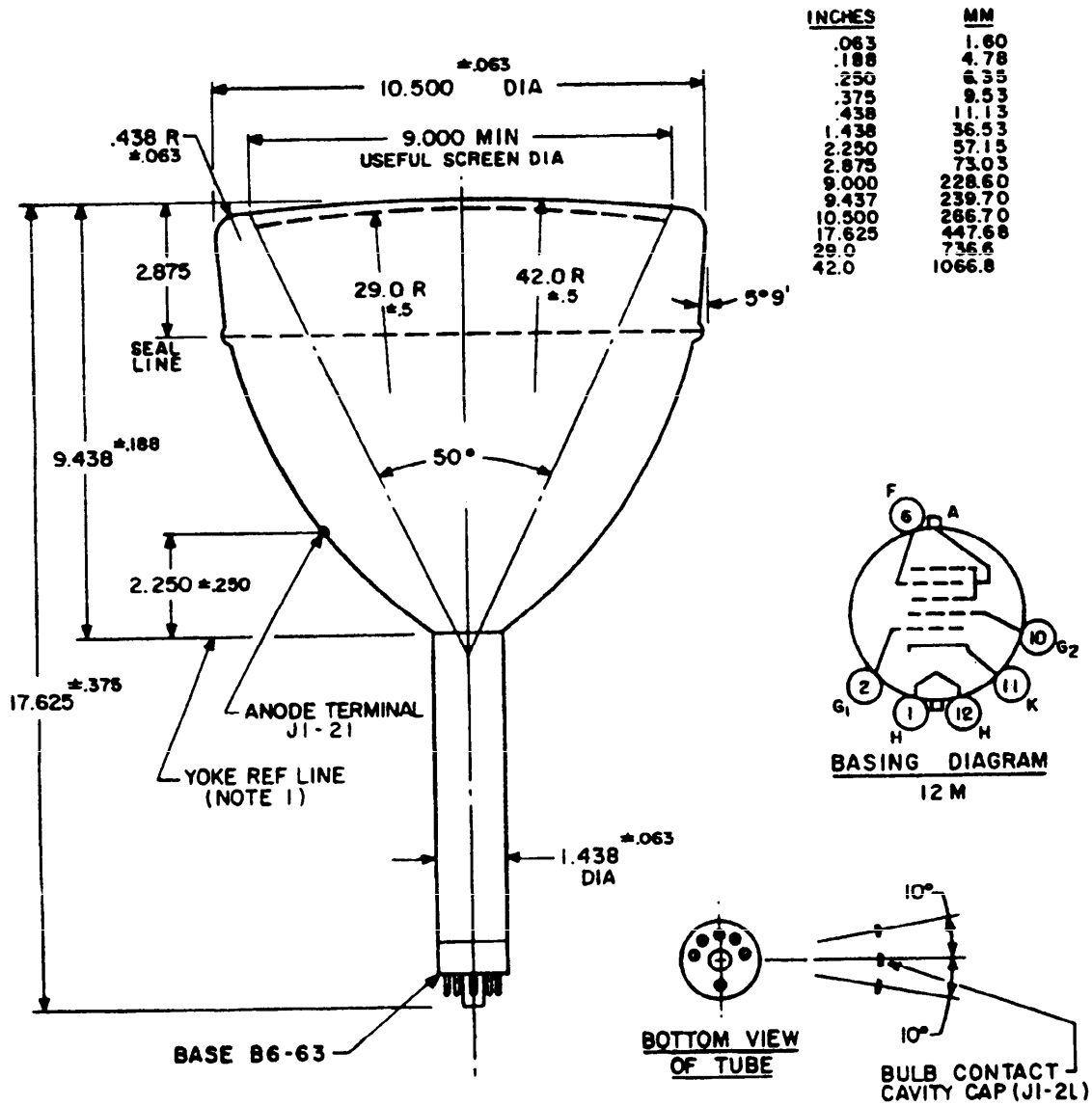


FIGURE 2 High Impedance Leakage

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:01Z
Check the source to verify that this is the current version before use.



Notes :

1) Reference line is determined by the plane of the of the upper edge of the reference line gage (EIA No. 112) when the gage is resting in the cone.
2) Face: Neutral density gray glass.
3) Dimensions in inches

FIGURE 1 10 UP 14 B Cathode Ray Tube

Source: https://assist.dla.mil Downloaded: 2026-10-15T14:01Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

C16061

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:01Z
Check the source to verify that this is the current version before use.

FOLD

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF THE AIR FORCE
DoD-318

U.S.MAIL

SMAMA/MMREE
Wright-Patterson AFB OH 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:01Z
Check the source to verify that this is the current version before use.